SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Case Study

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
```

```
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

ANSWERS: Web Development Standards Implementation

Quick summary of the correct choices:

Snippet	Requirement	Correct answer
1	Embed promotional video	B. <video src="video.mp4" controls></video>
2	Secure form submission	B. POST
3	Navigation menu (semantic)	B. <nav>
4	Valid XML syntax	B. <note><to>User</to><from>Admin</from></note>
5	Web communication protocol	B. HTTP (use HTTPS for secure transfer)

ANSWER 1: Correct HTML Tag for Video Embedding

Correct option: B. <video src="video.mp4" controls></video>

Justification

- **<video> is the standard HTML5 element** for embedding video. Browsers play the file natively, with no plugin such as Flash.
- The **controls** attribute shows the built-in play/pause, volume, seek bar and fullscreen buttons, so users can control the promotional video.
- The **src** attribute points to the video file (MP4 is supported by all major browsers).

- It offers useful extras: poster (preview image), width/height, autoplay with muted, loop, preload, and several <source> children for different formats.
- It supports fallback text and subtitles (<track>), which improves accessibility.

Why the other options are wrong

Option	Tag	Reason it is incorrect
A	<media>	Not an HTML element. Browsers ignore it and show nothing.
C	<movie>	Not an HTML element. It has no meaning to any browser.
D	<vid>	Not an HTML element. It is an invented tag and will not play the video.

Recommended code

```
<video src="video.mp4" controls width="640" height="360"
poster="thumbnail.jpg">
  Your browser does not support the video tag.
</video>
```

Conclusion: Option B is the only valid HTML5 video element, and it gives standards-compliant, plugin-free playback with user controls.

ANSWER 2: Appropriate HTTP Method for Form Submission

Correct option: B. POST

Justification

- **Data is sent in the request body**, not in the URL. Registration details such as name, email and password therefore do not appear in the address bar, browser history, bookmarks or server access logs.
- **No size limit** in practice, so long inputs and file uploads (multipart/form-data) work.
- **Not cached or bookmarked** by default, and the request will not be repeated accidentally.
- **Correct meaning:** POST is used to submit data that creates a new resource or changes server state, which is exactly what registering a user does.
- POST is **not encrypted by itself.** To send the data securely, the form must be served and submitted over **HTTPS**.

Why the other options are not suitable

Option	Method	Reason it is not suitable
A	GET	Appends data to the URL, so it is visible, logged, cached and bookmarkable. This is insecure for personal data and passwords.
C	PUT	Replaces or updates an existing resource at a known URL. HTML forms cannot submit PUT directly, and it is not meant for new registrations.
D	DELETE	Removes a resource. HTML forms do not support it, and it is the opposite of registering a user.

Recommended code

```
<form action="/register" method="POST">
  <input type="text" name="username" required>
  <input type="email" name="email" required>
  <input type="password" name="password" required>
  <button type="submit">Register</button>
</form>
```

> **Conclusion:** POST, sent over HTTPS, is the right method for a registration form because it keeps user data out of the URL and matches the purpose of creating a new record.

ANSWER 3: Proper Semantic Element for Navigation

Correct option: B. <nav>

Justification

- **<nav> is designed for major navigation links**, such as the main menu, a table of contents or pagination. Its name states its purpose.
- **Accessibility:** Screen readers treat <nav> as a navigation landmark, letting users jump straight to the menu.
- **SEO:** Search engines better understand the site structure and the importance of the links inside it.
- **Maintainability:** Code is easier to read and style than a page made only of generic tags.
- It meets the requirement to **follow semantic HTML standards** (HTML5).

Why the other options are not the best choice

Option	Element	Reason it is not the best choice
A		A generic container with no meaning. It does not tell browsers or assistive tools that the content is navigation.
C	<section>	Groups thematic content, usually with its own heading. It is not meant for menus.
D	<header>	Holds introductory content such as a logo and title. It may contain a <nav>, but it is not itself navigation.

Recommended code

```
<nav>
  <ul>
    <li><a href="index.html">Home</a></li>
    <li><a href="about.html">About</a></li>
    <li><a href="contact.html">Contact</a></li>
  </ul>
</nav>
```

> **Conclusion:** <nav> is the proper semantic element because it clearly marks the navigation menu for browsers, search engines and assistive technologies.

ANSWER 4: Correct XML Syntax

Correct option: B.

```
<note><to>User</to><from>Admin</from></note>
```

Why option B is correct

It is **well-formed**: it has a single root element (<note>), every start tag has a matching end tag, the tags are properly nested, and the tag names match exactly (XML is case-sensitive).

Errors in the other options

Option	Code	Error
A	<note><to>User</to><from>Admin</from>	The root element <note> is never closed. The closing </note> tag is missing.
C	<note><to>User</from></note>	Tags are mismatched: <to> is closed with </from>. The <to> element is never properly closed, and no <from> element is opened. The nesting is broken.
D	<note to="User"></note	The closing tag is incomplete: the final > is missing from </note. (The attribute itself is valid, since its value is in quotes.)

Basic rules of well-formed XML

- There must be exactly one root element.
- Every opening tag must have a matching closing tag.
- Elements must be properly nested and must not overlap.
- Tag names are case-sensitive.
- Attribute values must be enclosed in quotes.

> **Conclusion:** Only option B is well-formed XML. Options A, C and D each break a basic syntax rule (missing end tag, mismatched tag, and incomplete end tag), so an XML parser would reject them.

ANSWER 5: Correct Protocol for Web Communication

Correct option: B. HTTP (HyperText Transfer Protocol)

Justification

- **HTTP is the application-layer protocol of the World Wide Web.** Browsers and web servers use it to request and deliver web pages, images, scripts and data.
- It follows a **request-response model** over TCP (default port **80**), and uses methods (GET, POST), status codes (200, 404) and headers.
- The web application in the case (video, forms, pages) needs exactly this kind of communication.
- For security, the application should use **HTTPS** (HTTP over TLS/SSL, port **443**), which encrypts data such as registration details.

Why the other options are not suitable

Option	Protocol	Purpose (why it does not fit)
A	FTP	File Transfer Protocol: used to upload and download files between systems (ports 20/21). It does not deliver web pages to browsers.

Option	Protocol	Purpose (why it does not fit)
C	SMTP	Simple Mail Transfer Protocol: used to send email between mail servers (port 25/587), not for web content.
D	SNMP	Simple Network Management Protocol: used to monitor and manage network devices such as routers and switches (port 161).

> **Conclusion:** HTTP (preferably HTTPS) is the correct protocol because it is the standard protocol for browser-server web communication. FTP, SMTP and SNMP serve different purposes (file transfer, email and network management).

Code of Conduct:

I **K. Naga Venkata Siva / 192425345** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
K. Naga Venkata Siva
192425345

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly

Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand